

Documented prior art and limits of originality

Version 1.1.0 | Audited 8 October 2026 | Independent public-data research

Cash valuation and working capital

FCFF and finite-horizon discounting; accounting identities. Inspected scope: Existing baseline and financial mechanics. Overlap: No mathematical novelty; full cash/seasonal/replacement layer repairs baseline.

CVaR and robust decisions

Rockafellar-Uryasev CVaR; minimax regret. Inspected scope: CVAR original paper; six-world example. Overlap: Loss-tail and chosen regret applications; no fitted probabilities.

Causal commercial measurement

Callaway-Sant'Anna and randomized cluster inference. Inspected scope: DID original paper; NOBEL_2021 context. Overlap: Identification design only; actual pilot not executed.

Trade estimation

Santos Silva-Tenreyro PPML / OECD-FAO Aglink-Cosimo. Inspected scope: Official Outlook methodology and inspected PPML_METHOD author resource. Overlap: Appropriate conditional modules; not estimated from broad imports.

Adaptive learning

Bayesian EVSI/EVPI and staged decisions. Inspected scope: Two-state derivation and explicit study accuracy/cost. Overlap: Established Bayesian decision theory; no new theorem.

Network/portfolio

Constrained flow LP and dependency-aware allocation. Inspected scope: Executed scipy HiGHS flow and covariance $L L^T$. Overlap: Standard mathematical methods; no actual Olam synergy.

Commercial competitive integration

ADM/Cargill/Bunge-Viterra public disclosures. Inspected scope: Issuer pages inspected 8Oct2026. Overlap: Shows relevant service/cost/network competitive mechanisms; no exhaustive proprietary-system search.

What changes a decision in executed examples

Four indivisible actions: regret chooses partner and central NPV chooses owned. In an explicitly linear fractional relaxation, equal split lowers regret to 3.67 and the 27.27% owned/72.73% partner optimum lowers it to 2.91. These relaxations are not assumed feasible for actual lumpy assets; every policy shares the chosen 20 USDm aggregate envelope and payoff set. The imperfect-study example has gross EVSI 2.20, net EVSI 0.70 USDm at chosen prior/accuracy; capacity-marketing complementarity flips contribution sign. Model ablations use identical baseline inputs and expose cash/physical/learning mechanisms. These establish teaching counterexamples, not empirical Olam policy superiority.

Empirical proof still required

Compare equal allocation, market size, simple NPV, country-score and actual management heuristic against proposed policy on identical feasible projects, budget, horizon and information. Pre-register held-out decision regret, liquidity breaches, realized contribution, complexity and transition cost. Country-score and management benchmark are unexecuted because their criteria/data are unavailable. No exclusive mathematical novelty, independently validated optimal policy or exhaustive literature review is claimed.